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RULE · Simplifying (ratsional ifoda)

Factor fully, then cancel common **factors** only – never bare terms.

$$\frac{x^2-4}{x^2+5x+6}=\frac{(x-2)(x+2)}{(x+2)(x+3)}=\frac{x-2}{x+3}$$

$x \neq -2, -3$ – values that would make the original denominator 0 stay excluded even after canceling.

RULE · Multiplying & dividing

Factor first, then cancel. Dividing = multiplying by the reciprocal.

$$\frac{x^2-9}{2x}\times\frac{x}{x+3}=\frac{x-3}{2}$$

RULES SHEET – 2/2

RULE · Extraneous solutions (begona yechim)

Clear denominators, solve, then check the candidate in the **original** equation – reject it if any denominator becomes 0.

$$\frac{x}{x-2} + 1 = \frac{2}{x-2} \implies x = 2$$

Check: $x - 2 = 0$ at $x = 2$ – rejected. **No solution.**

Not every candidate is extraneous – always check, never assume.